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SUPPLEMENTARY MATERIAL

The Torsion Sector of the Tetrad Connection as the Regular Representation of S_4

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Abstract
The symmetric group S_4 permutes the four legs of a tetrad, and with them the 64 components of the affine connection. We show that the 24 torsion components form the regular representation of S_4, with an explicit generator: a single torsion tensor whose 24 images under the permutations of the legs are a basis. The proof splits the torsion components into two orbits of 12, according to whether the upper index repeats a lower one. The sign representation occurs once in the torsion sector and not at all in the 40-component Christoffel sector, and it is spanned by the totally antisymmetric torsion $\sum_d \varepsilon_{abcd}$. With indices lowered by $|\eta| = I$, the vector, axial and tensor parts of the torsion lie in the two halves as $12 = 4 + 8$ and $12 = 4 + 8$, and the two tensor halves are isomorphic; lowered by η itself, the axial part is not S_4-invariant. The regular representation fits the torsion sector only for $K = 2$ and $K = 4$ legs.

Setting
Let $V = \mathbb{R}^4$ have basis $f_0, \dots, f_3$, one vector per leg, and let S_4 act by $\sigma f_a = f_{\sigma(a)}$. Read by magnitude, the legs are equivalent: $|\eta_{ab}| = \delta_{ab}$ is invariant under every permutation [1]. The dual basis is permuted in the same way, so $V^* \cong V$ as S_4-modules with no metric. The affine connection $\Gamma^a{}_{bc}$ lies in $V \otimes V^* \otimes V^* \cong V \otimes V \otimes V$, which splits into the Christoffel sector $C = V \otimes \mathrm{Sym}^2 V$ (40 components) and the torsion sector $T = V \otimes \Lambda^2 V$ (24 components), both S_4-submodules [2]. The torsion tensor takes values in T in any basis; in a coordinate basis, C and T are the symmetric and antisymmetric parts of Γ. T has basis $t_{a;bc} = f_a \otimes f_b \wedge f_c$, $b < c$, and $\sigma\, t_{a;bc} = t_{\sigma(a);\sigma(b)\sigma(c)}$, with $t_{a;cb} = -t_{a;bc}$.

We write $\mathbf{1}$, sgn, $[2,2]$, std and sgn $\otimes$ std for the irreducible representations of S_4, of dimensions $1, 1, 2, 3, 3$ [4], and $H = \langle (23) \rangle$.

Theorems 1 and 2 and the Proposition use only the labels of the legs; Theorem 3 lowers an index with $|\eta| = I$.

Two Orbits
Lemma. *Up to sign, the basis vectors of T form two S_4-orbits of 12: the incident vectors, with $a \in \{b, c\}$, and the non-incident vectors, with $a \notin \{b, c\}$. The stabiliser of the line of $t_{0;01}$, and of the line of $t_{0;23}$, is H; it acts by $+1$ on $t_{0;01}$ and by -1 on $t_{0;23}$.*

Proof. An incident vector is determined by the ordered pair (a, b) of distinct labels with $\{b, c\} = \{a, b\}$, and a non-incident vector by the ordered pair (a, d), where d is the label outside $\{a, b, c\}$. S_4 is transitive on the 12 ordered pairs of distinct labels, so each family is one orbit. A permutation fixing the line of $t_{0;01}$ fixes 0 and the set $\{0, 1\}$, hence 1, and is e or (23); $(23)\, t_{0;01} = t_{0;01}$. One fixing the line of $t_{0;23}$ fixes 0 and the set $\{2, 3\}$, hence 1, and is e or (23); $(23)\, t_{0;23} = t_{0;32} = -t_{0;23}$. □

The Regular Representation
Theorem 1. *Let $\xi = t_{0;01} + t_{0;23}$. The 24 vectors $\sigma\xi$, $\sigma \in S_4$, are a basis of T, and*

$$\Phi : \mathbb{R}[S_4] \to T, \qquad \sigma \mapsto \sigma\xi, \tag{1}$$

is an isomorphism of S_4-modules: the torsion sector is the regular representation.

Proof. With $\tau = (23)$, the Lemma gives $\tau\xi = t_{0;01} - t_{0;23}$. Hence, for every σ, $\sigma\xi + \sigma\tau\xi = 2\,\sigma t_{0;01}$ and $\sigma\xi - \sigma\tau\xi = 2\,\sigma t_{0;23}$. As σ runs over S_4, these run over all incident and all non-incident basis vectors up to sign, so the 24 vectors $\sigma\xi$ span T, which has dimension 24; they are a basis. Φ sends the basis $\{\sigma\}$ of $\mathbb{R}[S_4]$ to this basis, and $\Phi(\rho\sigma) = \rho\sigma\xi = \rho\,\Phi(\sigma)$. □

Corollary 1. *Each irreducible representation occurs in T with multiplicity equal to its dimension,*

$$T \cong \mathbf{1} \oplus \mathrm{sgn} \oplus 2\,[2,2] \oplus 3\,\mathrm{std} \oplus 3\,(\mathrm{sgn} \otimes \mathrm{std}), \tag{2}$$

and $T = T_{\mathrm{inc}} \oplus T_{\mathrm{non}}$ with $T_{\mathrm{inc}} \cong \mathrm{Ind}_H^{S_4} \mathbf{1}$ and $T_{\mathrm{non}} \cong \mathrm{Ind}_H^{S_4} \varepsilon$, where ε is the sign of H.

Proof. The first statement holds for every regular representation [3]. A module spanned by one orbit of lines, on which the stabiliser H of one line acts by a character χ, is $\mathrm{Ind}_H^{S_4} \chi$ [3]; apply the Lemma. As a check, $\mathrm{Ind}_H(\mathbf{1} \oplus \varepsilon) = \mathrm{Ind}_H \mathbb{R}[H] = \mathbb{R}[S_4]$ by transitivity of induction, and counting fixed lines with their signs gives the character of T as $(24, 0, 0, 0, 0)$ on the classes e, (ab), $(ab)(cd)$, (abc), $(abcd)$. □

The Sign Representation

Theorem 2. sgn *occurs exactly once in* T *and not at all in* C*. In* T *it is spanned by* ω*, with components* $\omega^a{}_{bc} = \sum_d \varepsilon_{abcd}$*, which lies in* T_{non}*.*

Proof. By Frobenius reciprocity [3], sgn occurs in $\mathrm{Ind}_H \chi$ once if $\chi = \mathrm{sgn}|_H$ and otherwise not. Since $\mathrm{sgn}|_H = \varepsilon$, it occurs once in T_{non} and not in T_{inc}. The basis $f_a \otimes f_b f_c$ of C is permuted without signs, in four orbits: $a = b = c$ (4), $b = c \neq a$ (12), $a \in \{b, c\}$ with $b \neq c$ (12), and a, b, c distinct (12). Each stabiliser contains a transposition acting by $+1$, so C is a sum of modules $\mathrm{Ind}_{H_i} \mathbf{1}$ with $\mathrm{sgn}|_{H_i} \neq \mathbf{1}$, none of which contains sgn. Finally $\varepsilon_{\sigma(a)\sigma(b)\sigma(c)\sigma(d)} = \mathrm{sgn}(\sigma)\, \varepsilon_{abcd}$ and the sum over d is invariant, so $\sigma\omega = \mathrm{sgn}(\sigma)\, \omega$; $\omega \neq 0$ is antisymmetric in b, c and vanishes unless a, b, c are distinct. □

Vector, Axial and Tensor Parts

Lower indices with $|\eta| = I$. The torsion splits into its vector part $T_v = \{\delta^a{}_b v_c - \delta^a{}_c v_b\}$, its axial part $T_{ax} = \{\varepsilon_{abcd}\, w^d\}$, both of dimension 4, and its tensor part T_t, of dimension 16, the torsion that is trace-free and has no totally antisymmetric part [2].

Theorem 3. $T_v \subset T_{\text{inc}}$ *and* $T_{ax} \subset T_{\text{non}}$*, and* $T_t = (T_{\text{inc}} \ominus T_v) \oplus (T_{\text{non}} \ominus T_{ax})$*. As* S_4*-modules*

$$\begin{aligned} T_v &\cong V = \mathbf{1} \oplus \mathrm{std}, \\ T_{ax} &\cong \mathrm{sgn} \otimes V = \mathrm{sgn} \oplus (\mathrm{sgn} \otimes \mathrm{std}), \end{aligned} \tag{3}$$

and each half of T_t *is* $[2,2] \oplus \mathrm{std} \oplus (\mathrm{sgn} \otimes \mathrm{std})$*. Lowered by* η *itself, the axial part is not* S_4*-invariant.*

Proof. The components of T_v vanish unless $a \in \{b, c\}$, and those of T_{ax} unless a, b, c are distinct. The trace $T^a{}_{ac}$ involves only incident components, and T_v is the image of its adjoint, so $T_{\text{inc}} \ominus T_v$ is the trace-free part of T_{inc} and has no totally antisymmetric part. Likewise T_{ax} is the image of total antisymmetrisation, which involves only non-incident components, so $T_{\text{non}} \ominus T_{ax}$ is trace-free with no totally antisymmetric part. Both halves lie in T_t, and their dimensions sum to $16 = \dim T_t$. The maps $v \mapsto T_v$ and $w \mapsto T_{ax}$ are equivariant, the second with the sign of ε. Counting fixed lines with signs gives $\chi_{\text{inc}} = (12, 2, 0, 0, 0)$ and $\chi_{\text{non}} = (12, -2, 0, 0, 0)$; subtracting $\chi_V = (4, 2, 0, 1, 0)$ and $\mathrm{sgn}\, \chi_V$ leaves $(8, 0, 0, -1, 0)$ for each half, the character of $[2,2] \oplus \mathrm{std} \oplus (\mathrm{sgn} \otimes \mathrm{std})$. With η, the axial tensor whose η-lowered components are ε_{abc2} is sent by (01) to one with lowered components $T_{013} = -1$ and $T_{301} = +1$, which is not totally antisymmetric. □

Why Four Legs

Proposition. *For* K *legs,* $\dim T = K\binom{K}{2}$ *equals* $|S_K| = K!$ *only for* $K = 2$ *and* $K = 4$*, and at both the torsion sector is the regular representation.*

Proof. For $K \geq 2$ the equation reduces to $K/2 = (K-2)!$. It holds at $K = 2$ and $K = 4$, fails at $K = 3$, and for $K \geq 5$, $(K-2)! \geq (K-2)(K-3) \geq 2(K-2) > K/2$. At $K = 1$, $\dim T = 0 \neq 1$. At $K = 2$, T is spanned by $t_{0;01}$ and $t_{1;01}$, which the transposition exchanges with a sign, so its character is $(2, 0)$; $K = 4$ is Theorem 1. □

Scope

$\mathrm{sgn}(\sigma) = \det P_\sigma$: the sign representation is the orientation character of $GL(V)$ restricted to permutations of the legs. Spatial parity, $\mathrm{diag}(1, -1, -1, -1)$, also reverses orientation, but it is not a permutation of the legs, and these theorems do not describe its action. Theorems 1 and 2 hold for any four-index frame, with or without a metric. Theorem 3 uses the magnitude reading $|\eta| = I$; with the sign of η kept, the axial part is not S_4-invariant.

Summary

The 24 torsion components of the tetrad connection are one copy of the group algebra of the legs' permutations, generated by one explicit tensor. The sign representation lives in the torsion alone, as its totally antisymmetric part along $(1, 1, 1, 1)$. Read by magnitude, the vector part of the torsion is incident, the axial part is non-incident, and the tensor part is two isomorphic halves. General relativity's torsion-free condition therefore removes a complete copy of $\mathbb{R}[S_4]$, and with it the only scalar response of the connection to odd permutations of the legs.

References

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